Interference

Ghatak §13.5, 14.1, 14.3–14.12, 15.1–15.4, 15.8–15.11 · pattern of the Nov 2024 / Nov 2025 papers, doubled

HOW THIS PAPER WORKS

A real paper has 12 short, 7 paragraph and 2 essay questions. This paper has **twice as many** (24 + 14 + 4), all from Module 2, covering every concept once as its main point.

Time	4 hours (or two sittings of 2 hours)
Maximum	120 marks

PYQ N25 Q3 = Nov 2025, question 3. "also" lists other years that asked the same thing. "adapted" = a 1-mark fill-in or True/False turned into a short answer, or two past questions joined.

Section A · Short answer

Answer all questions in two or three sentences. Each carries 2 marks. Ceiling 40.

1. State and explain the principle of superposition of light waves.

PYQ N19 Q15 **also N24 Q4**

2. Does intensities of waves obey superposition principle? Justify your answer.

PYQ N25 Q2

3. Two monochromatic waves of same intensity I and same wavelength λ interfere at a point. If the two waves are completely incoherent, what will be the resultant intensity?

PYQ N25 Q3

4. Explain coherence.

PYQ N24 Q5

5. Describe the conditions to obtain sustained interference pattern.

PYQ N22 Q13

6. State the principle behind the interference obtained by division of wave front and amplitude methods.

PYQ N20 Q13

7. What is coherence length? Estimate it for light whose wave trains last about 10^{-10} s.

8. In Young's experiment, λ = 500 nm, the slits are 0.5 cm apart and the screen is 1 m away. Find the bandwidth. Is it true that, to observe a good interference pattern, the screen must be close to the sources?

PYQ N25 Q14 + N22 Q7 · adapted

9. Define the visibility of fringes. Find it when one interfering beam is four times as intense as the other.

10. What is the exact shape of the interference fringes of two point sources on a plane screen? Why do they look straight?

11. Young's experiment is done with white light. Describe the pattern. Why is the central fringe white?

12. In Fresnel's two-mirror arrangement, why are the two virtual sources coherent? What is their separation?

13. In Lloyd's mirror the fringe at the edge of the mirror is dark. What does this show?

14. "The optical path traversed by light wave when reflected from the lower surface of a thin film ____ with increase of refractive index of the film." Fill in and explain.

PYQ N19 Q5 · adapted

15. For the normal incidence of a parallel light beam on a thin film of refractive index n, state the condition for destructive and constructive interference. Explain. How do the conditions change if the film's index lies between those of the media on either side?

PYQ N19 Q13

16. Light of wavelength 5839 Å is reflected at near normal incidence from a soap bubble of refractive index 1.42. What is the least thickness of the film that will appear bright by reflection?

PYQ N22 Q27

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17. Explain colour of thin films.

PYQ N19 Q14 **also N22 Q14 · N24 Q7 · N22 Q8**

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18. Interference colours are not observed in thick plates or glass slabs. Why?

PYQ N19 Q17

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19. "Lenses are coated with non-reflecting films to minimise ____." Fill in. Why does a coated lens look purple?

PYQ N22 Q4 · adapted

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20. "The fringes obtained from two perfectly plane surfaces are not equal." True or false? Explain.

PYQ N20 Q10 · adapted

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21. What is the central spot of Newton's rings in reflected light, and in transmitted light? Why?

PYQ N19 Q3 · adapted **also N20 Q3**

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22. Why do Newton's rings get closer together away from the centre? What happens to them when a liquid is put between the lens and the plate?

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23. State the condition for constructive interference for a Michelson interferometer and explain the terms therein.

PYQ N25 Q4

24. Why is a compensating plate used in a Michelson interferometer?

Section B · Paragraph / problem

Answer all questions in about half a page to one page. Each carries 5 marks. Ceiling 60.

25. Considering the super position of two sinusoidal waves, obtain the conditions for constructive and destructive interference.

PYQ N22 Q19

26. Show that a thin plate of thickness t and index n placed over one slit in Young's experiment shifts the fringes by $D(n-1)t/d$. A mica sheet (n = 1.58) shifts the central fringe by 0.2 cm, with d = 0.1 cm and D = 50 cm. Find its thickness.

27. Explain Fresnel's biprism, and how it is used to determine the wavelength of light.

PYQ N20 Q19

28. In a biprism experiment, interference fringes are observed at a distance of 1 m away from it. The refractive index of the material of the biprism is 1.5. If the distance between the source and the biprism is 10 cm, calculate the fringe width when wavelength of light used is 5000 Å. Given that the biprism has refracting angle 1°.

PYQ N19 Q26

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29. In an interference experiment, the slit source is at a distance of 2 mm from the plane of the mirror. If the wavelength of the light used is 589 nm and the screen is kept at a distance of 1.5 m from the source, calculate the fringe width. Describe the arrangement.

PYQ N22 Q26

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30. Using the principle of optical reversibility, derive the Stokes relations. Hence explain the phase change on reflection.

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31. Explain the interference by a plane parallel film when illuminated by a plane wave.

PYQ N24 Q13

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32. Two plane waves of equal amplitude travel at angles +θ and −θ to the z-axis. Show that the intensity is $4I_0 \cos^2(kx \sin\theta)$ and find the fringe width. Evaluate it for θ = 30°, λ = 5000 Å.

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33. With necessary figure, explain interference by a plane parallel film illuminated by a point source.

PYQ N24 Q16 | also N20 Q20

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34. Write a note on non reflecting films.

PYQ N24 Q15 | also N19 Q22

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35. Two plane glass plates are placed on top of each other, and on one side a cardboard is introduced to form a thin wedge of air. Assuming that a beam of wavelength 6000 Å is incident normally, and there are 100 interference fringes per centimetre, calculate the wedge angle.

PYQ N20 Q31

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36. Newton's rings are formed with red light of λ = 670 nm. The radius of the 20th dark ring is found to be 1.1×10^{-2} m. Find the radius of curvature of the lens and the radius of the 30th ring.

PYQ N22 Q28 | also N19 Q27 · N20 Q26

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37. Newton's rings are formed with sodium light (5890 Å and 5896 Å) and a lens of radius of curvature 100 cm. At what thickness of the air film do the rings first become indistinct? Near which ring and at what radius does this happen?

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38. With neat diagram, explain Michelson interferometer. Describe how it can be used to determine the difference in wavelength of two waves.

PYQ N22 Q20

Section C · Essay

Answer any two, in about two pages each. Each carries 10 marks.

39. What are coherent sources? Derive an expression for the resultant intensity when two coherent beams of light are superposed, and show that the fringes are equally spaced. What will be the intensity when two sources are incoherent?

PYQ N19 Q32 · also N24 Q20

40. Obtain the conditions for maximum and minimum of interference pattern on a parallel thin film (derive the cosine law), in reflected and in transmitted light.

PYQ N25 Q20 · also N24 Q14

41. Explain with necessary theory, the formation of Newton's rings in reflected light. How can we use this arrangement to determine the wavelength of light?

PYQ N22 Q32

42. Describe Michelson interferometer and explain how it is used to standardize the meter. Explain how circular, straight and white light fringes are formed.

PYQ N20 Q32 **also N19 Q21**

Coverage check

TOPIC	QUESTIONS
T1 Superposition	1, 2, 25
T2 Coherence	3, 4, 5, 6, 7
T3 Young: path, fringe width, shape	8, 10, 39
T4 Intensity, visibility	9, 39
T5 Plane waves, wide source	32
T6 Fringe shift	26
T7 Two mirrors	12
T8 Biprism	27, 28
T9 White-light fringes	11
T10 Lloyd's mirror	13, 29
T11 Phase change, Stokes	13, 30
T12 Film, normal incidence	14, 15, 16
T13 Cosine law, plane wave	31, 40
T14 Point source, Haidinger	33
T15 Non-reflecting films	19, 34
T16 Colours, thick plates	17, 18
T17 Wedge	20, 35
T18 Newton's rings	21, 22, 36, 41
T19 Two-λ rings	37
T20 Michelson	23, 24, 42
T21 Michelson Δλ	38

All 43 Module 2 past questions (2019–2025) appear, directly or as "also".